

Though a truly comprehensive analysis would include these sorts of risk and term premia, they will not be considered explicitly in this book, for two reasons. First, these premia are not observable *ex ante*, and a comprehensive treatment of these premia is simply beyond the scope of this book.

Second, this book deals primarily with the systematic difference between interest rates along three yield curves with maturities of up to fifty years in some cases. It's unlikely (though not inconceivable) that these risk and term premia could explain temporary anomalies between spreads in isolated and long-term segments of these yield curves.

The Effect of Default Risk

Perhaps the most important reason for the systematic difference between LIBOR and OIS involves default risk. The OIS rate is the interbank rate for overnight lending between banks, all of which are healthy on the day the transaction is initiated. The scope for a bank to transition overnight from being healthy to being bankrupt is fairly limited. In contrast, there is greater scope for a healthy bank to transition to bankruptcy over, say, six months, a common tenor for LIBOR. By using OIS rather than term lending, the lender can reassess the credit quality of his counterpart every day and in case of emerging signs of stress immediately select a different borrower for the next day. This ability greatly reduces the credit exposure even for unsecured rates.

As this concept is particularly important, we'll reiterate the importance of these credit 'refreshes.' If an investor lends at the 6M interbank deposit rate, he subjects his deposit to the possibility that the borrowing bank will default at some point during those six months, and his deposit rate reflects that risk. If, as an alternative, he sticks with one-month deposits, which he rolls five times after the initial deposit, then he subjects his deposit to less default risk, since an institution whose credit deteriorates after, say, three months can be avoided when rolling the deposit in the latter scenario, while our investor would be subject to this credit deterioration in the former scenario.

In other words, 6M LIBOR reflects the risk that institutions in the LIBOR fixing panel might default during those six months. But by dealing with LIBORs of lesser tenors, we limit the extent of default risk, as subsequent LIBOR fixings are in some sense 'refreshed' by the fact that poor credits are removed from the fixing panel. So exposure to two 3M LIBORs in sequence will expose a trader to less credit risk than will a single 6M LIBOR. And a series of overnight index swaps will subject the trader to far less credit risk, as each overnight rate in the sequences exposes the trader only to one day's worth of credit risk.

The implication here is quite important. As credit conditions in the banking sector deteriorate, LIBORs should increase relative to OIS rates, all things being equal. And, of course, this is precisely the behavior observed in practice.

COMBINING THE DRIVING FORCES CAPITAL REQUIREMENTS AND CREDIT EXPOSURE INTO A MODEL FOR THE REPO–LIBOR SPREAD

This chapter started with a classification of the reference rates, before assessing the most important driving forces of the spreads between them: the cost of capital for the spread between secured (repo) and unsecured (Fed Funds, ESTR) O/N rates, the credit exposure for the spread between unsecured O/N and term rates (LIBOR). Combining both, we can expand the simple formula from above ($g*q*d$) with credit terms to a model for the repo–LIBOR spread, i.e. between the third and first column of Table 11.1. This will be the basis for pricing asset swap spreads of bonds versus LIBOR in Chapter 12: the funding difference and the credit exposure of the floating leg are already covered by the model, hence only the credit exposure of the fixed leg (the sovereign issuer) still needs to be added.

Assume a bank⁹ wants to make a loan for a single period. If the loan is to be secured via repo collateral, the pre-tax return at the end of the period will be $[1 + R]$, where R is the repo rate for this single period. Assuming the loan is collateralized with a high-quality government bond, the bank can use a low-cost source of funds (e.g. bank retail deposits) to provide for the loan. If the cost of this capital is given by b , the cost of the loan is $[1 + b]$. The profit on this transaction is then $[1 + R] - [1 + b]$.

If the bank makes an unsecured interbank loan instead, it will have to finance dq of this loan at its higher cost of equity capital, g , where q is the BIS risk-weighting for interbank loans and d is the BIS guideline for the capital adequacy ratio for a 100% risk-weighted asset (typically 8%). The cost of capital will then be given by the expression:

$$(1 - qd)[1 + b] + qd[1 + g]$$

⁹This model is constructed from the perspective of a bank lending cash. We ignore any possible impact of haircuts for simplicity at this point. From the perspective of the bank, this would be valid, for example, in a case in which the loan was over-collateralized with government bonds, with the borrower of cash receiving any interest payments on the bonds.

Now, we add the credit component. If the probability of default is given by p , and the recovery percentage is given by c (percentage of terminal liability recovered rather than percentage of the current market value), then the expected return of the interbank loan L is given by:

$$pc[1 + L] + (1 - p)[1 + L]$$

Then the expected profit is given by:

$$pc[1 + L] + (1 - p)[1 + L] - (1 - qd)[1 + b] - qd[1 + g]$$

Assuming that the default probability is a risk-neutral default probability (i.e. it already incorporates the effect of any covariance-related risk premia), the expected profits of the secured and unsecured transactions can be equated to obtain an expression relating LIBOR and the repo rate. In particular:

$$[1 + R] - [1 + b] = pc[1 + L] + (1 - p)[1 + L] - (1 - qd)[1 + b] - qd[1 + g]$$

This equation can be solved in terms of LIBOR.

$$L = \frac{R}{1 - p(1 - c)} + \frac{p(1 - c)}{1 - p(1 - c)} + \frac{qd(g - b)}{1 - p(1 - c)}$$

The first term on the right-hand side of this equation is an amount to act as insurance against the default on the interest payment. In the degenerate case that $R = 0$, there is no interest payment against which to insure, and this term equals zero. And in the case that the recovery rate in the event of default is 100% (again, as a percentage of terminal value rather than nominal value), then the denominator in the first term equals 1, in which case this term contributes nothing to the repo-LIBOR spread. And if $p = 0$, so there is no probability of default, then the denominator of this expression is also 1, and this first term contributes nothing to the spread.

Note also that as long as there is some probability of default ($p > 0$) and the recovery value, c , is less than 100%, the denominator of this first term is less than 1, so that LIBOR increases at a rate faster than the repo rate. As a result, the difference between LIBOR and repo in general should be an increasing function of interest rates, all things being equal.

The second term on the right-hand side of the equation is an amount to insure against the principal of the loan. Note that this term equals zero if the recovery value is 100% ($c = 1$) or if the default probability is zero ($p = 0$).

The third and final term on the right-hand side of the expression reflects the additional amount that needs to be charged to compensate for the higher cost of equity capital on the portion of the unsecured loan that needs to be funded with core capital to satisfy the bank's regulatory capital requirement. If the cost of equity capital, g , happens to be the same as the marginal cost of borrowed funds, b , then this term is zero. Likewise, if the regulatory risk weighting for the unsecured loan is zero ($g = 0$) or if the capital adequacy ratio is zero ($d = 0$), then this term is also zero.

Overall, this equation explains and quantifies the empirically observed behavior of the difference between LIBOR and repo ($L - R$), which is

- an increasing function of the level of interest rates;
- an increasing function of the cost of core capital;
- an increasing function of required capital ratios.

Asset Swaps

GENERAL CONCEPT

When looking at investing in a government bond from the viewpoint of a derivative, it appears to be a swap between the coupon as fixed rate and the repo as floating rate: until the maturity of the bond, one can finance the investment by using the bond as collateral in the repo market, i.e. by paying its repo rate (e.g. rolling O/N). Hence, investing in a bond and funding the position until maturity in the repo market is in principle the same as a pay repo–receive coupon swap (whose tenor is given by the maturity of the bond).

This derivative-like perspective on a bond makes it easy to compare bonds with swaps: Imagine that at the same time as investing in the bond we enter into a swap agreement paying fixed and receiving floating. If we choose the dates and size of the fixed payments in the swap to match the coupon dates and size of the bond – which is what happens in an asset swap – the fixed payments cancel out.¹ Thus, the combination of a bond and a swap in an asset swap results in a pay repo of the bond versus receive the floating rate of the swap position. In case the floating rate of the swap is LIBOR, an asset swap can therefore conceptually be thought of as a repo–LIBOR basis swap over the life of the bond. And the asset swap spread, i.e. the amount added to the floating rate of the swap position,² can be determined (in part) by the fair value of a repo–LIBOR basis swap. In other words, the asset swap spread includes the market price compensating for the privilege to fund at repo rather than LIBOR. While these basis swaps do not trade, we can price them via the model

¹In order to focus on the big conceptual picture, we need to abstract from several complicating features, such as bonds not trading at par, different daycount conventions, etc. These features will be added subsequently.

²This definition results in the convention Reference Rate + X for the asset swap spread X. For example, if a Bund quotes at EURIBOR –20 bp, its asset swap spread is –20 bp (not +20bp). Under this convention, a relative richening of bonds versus swaps corresponds to decreasing asset swap spreads. In line with this convention, the funding difference of asset swap spreads versus LIBOR is given by a repo–LIBOR (not LIBOR–repo) basis swap.

presented in Chapter 11 and thereby obtain a starting point for pricing asset swap spreads: the fair value of the repo–LIBOR basis swap, using the anticipated repo rates and maturity of a specific bond, is one component of the fair value of the asset swap spread of that bond.

This is a conceptual approach³ to asset swap spreads, which directly and causally captures the different funding rates for bonds (repo) and swaps and allows the application of cash-flow arbitrage principles to asset swap spread pricing: since an asset swap matches the fixed payments of the bond and swap, the asset swap spread can be determined as fair value of the basis swap between the two floating rates, i.e. repo and the reference rate used in the floating leg of the swap.

Apart from exchanging funding rates, an asset swap also exchanges credit exposure: investing in a bond exposes one to sovereign default risk, while a swap with LIBOR as reference rate exposes one to bank default risk. In other words, an asset swap matches the fixed cash flows and exchanges the floating cash flows, but these cash flows are subject to different credit risk.

Adding the credit perspective to the funding perspective from above, we can compare the two sides of an asset swap, choosing LIBOR as example for the reference rate:

- The holder of the bond is subject to the default risk of the issuer. In case of a sovereign default, despite losing the principal, he still needs to serve his fixed payment obligations in the asset swap, but will not receive the matching coupon payments from the bond anymore.
- If he takes the other side of the asset swap, i.e. he will receive fixed/pay floating, he needs to generate the floating payments, i.e. LIBOR, by lending unsecured to (LIBOR panel) banks. Again, in case of the borrower defaulting, despite losing the principal, he still needs to serve his floating payment obligations in the asset swap, but will not receive the matching LIBOR payments anymore.

Altogether, one can consider an asset swap as a combination of a funding swap and a credit swap:

- An exchange between the funding rate of the bond (repo) and the reference rate of the swap, i.e. a repo–swap reference rate basis swap over the life of the bond
- An exchange between the credit exposure to the bond issuer and to the counterparties in the money market.

³In the first edition of this book, we compared this “modern” with the “old” approach. By now, this is more of historic interest and has therefore been omitted from the second edition.

As argued in Chapter 11, the credit risk to counterparties in the money market is usually negligible for rolling O/N loans. Note that the credit ‘refreshes’ in the money market are a significantly different feature from the credit exposure of a bond investment: buying and holding a 10Y government bond expose one to the same counterparty for the next decade. Hence, the credit exposure on the floating side is in practice usually only relevant for swaps with LIBOR as the reference rate. And as it has already been incorporated into the model for the repo–LIBOR spread from Chapter 11, this model covers both the funding aspects and the credit aspects of the floating side.

With an asset swap combining a funding and a credit swap, it can be priced by expanding the funding model with a credit component:

- The model for the spread between different reference rates gives a fair value for the repo–swap reference rate basis swap, i.e. the *funding component* of swap spreads.
- Adding the *credit component* of swap spreads on the *floating side* only needs to be done in case of LIBOR as reference rate and has already been done in the repo–LIBOR model at the end of Chapter 11.
- Hence, only the *credit component* of swap spreads on the *fixed side*, i.e. the exposure to sovereign default risk, still needs to be added to the model.

If the credit default swap (CDS) was a pure measure of default risk, one could complete the task of building a pricing model for asset swap spreads by simply adding it to the fair value of the repo–swap reference rate basis swap, i.e.

Asset swap spread = Fair value of the basis swap between repo and the reference rate of the swap (from model) + CDS (from market),

in the case of LIBOR thus:

Asset swap spread = repo–LIBOR basis swap (from model) + CDS (from market).

Unfortunately, the elegance of this concept is tainted by the fact that the CDS is not a pure measure of default risk and therefore requires adjustments before being used in the formula above. This is a complex topic, which occupies Chapter 13. Previewing the results from Chapter 13 (and keeping the caveats under which they are derived in mind), the formula will read:

*Asset swap spread = Fair value of the basis swap between repo and the reference rate of the swap (from model) + Adjusted CDS, with Adjusted CDS = CDS (from market) * 0.45 for cash-settled CDS and Adjusted CDS = CDS (from market) * 0.39 for CDS with physical delivery.*

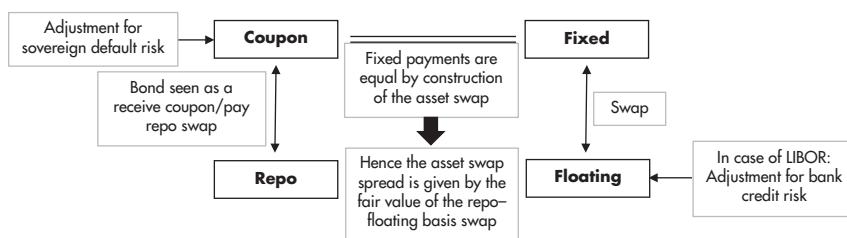


FIGURE 12.1 Conceptual approach to asset swap pricing.

Source: Authors.

Before applying the general concept to the different types of reference rates and discussing further additions, it may be useful to reflect on its fundamental principles. On a very basic level, we have considered only the cash flows of an asset swap. While this is the precondition to apply no-arbitrage principles, it also excludes features not accessible via cash flow analysis from the onset. These features, for example, the regulatory benefits of holding bonds, therefore need to be treated separately in Chapter 18, after the cash flow-based models will have been completed. Looking at the cash flows of an asset swap and thinking of a bond like a swap between fixed (coupon income) and floating (repo financing), we applied a no-arbitrage principle by arguing, that as the fixed cash flows match by construction of the asset swap, also the floating cash flows should match, hence that the asset swap spread should be equal to the fair value of the repo–swap reference rate basis swap over the life of the bond. This argument is illustrated by the fat arrow in Figure 12.1. Having captured the funding element of asset swaps, we have then turned our attention to the different credit exposure and found that while the cash flows match in case of no default, they are subject to different default risk. In order to maintain the cash flow-matching in the presence of default risk, corresponding adjustments are required: the adjustment for the credit exposure to LIBOR has already been incorporated in the model for the repo–LIBOR spread, while the adjustment for the credit exposure to bond issuers can be done by adding the CDS (subject to the modifications in Chapter 13).

APPLICATION OF THE GENERAL CONCEPT TO PRICE THE THREE TYPES OF ASSET SWAPS

As discussed in Chapter 11, there are three types of swaps, depending on the reference rate used on the floating side and therefore also three types of asset swap spreads, all of which can be considered as special cases of the general concept and which will now be treated individually.

Swap Spreads Versus SOFR

In case of the floating rate of the asset swap being SOFR, the asset swap spread (“ASW”) of a bond versus SOFR (the “SOFR–ASW”) is given by the following special case of the general formula above:

$$\text{SOFR–ASW} = \text{repo–SOFR basis swap} + \text{adjusted CDS}.$$

As mentioned in Chapter 11, due to the calculation process of SOFR, most special repo rates are excluded. Hence, SOFR is generally close to the GC rate. Still, deviations between SOFR and GC can arise from several sources, including:

- Since only FICC transactions below the volume-weighted 25th percentile are removed, in case more repo trades at special rates take place during a certain day, some of them enter the SOFR calculation.
- Due to the additional constraint $\text{SOFR} \leq \text{SRFR}$, SOFR could be lower than GC transactions in the actual market. This difference is likely to be mitigated by the SRFR influencing the repo market – as long as there is enough arbitrage capital available.

If we assume – subject to the caveats just mentioned – $\text{SOFR} = \text{GC}$, then the repo–SOFR basis swap equals a repo–GC basis swap, i.e. a daily exchange between the repo rate of the specific bond used in the asset swap and the GC rate. This is precisely the specialness of the bond and the formula is thus

$$\text{SOFR–ASW} = \text{repo–GC basis swap (specialness)} + \text{adjusted CDS}.$$

In case of SOFR–ASW and under the assumption $\text{SOFR} = \text{GC}$, the funding element of the asset swap is therefore simply the specialness of the bond, and the fair value of the SOFR–ASW is given by the expected specialness of the bond over its entire life (plus the adjusted CDS). In other words, a SOFR–ASW exchanges every day the repo rate of a specific bond versus the GC rate. In case of the bond trading GC, the value of this exchange is zero. In case of the bond trading special, the spread to GC, i.e. its specialness, is exchanged. Hence, the fair value of the repo–GC basis swap used in the ASW formula is given by the sum of the discounted (O/N-)specialness of the bond for every day of its remaining life.

One can estimate the evolution of specialness over the life of a bond by looking at its predecessors. For example, Bunds trade typically special as 10Y benchmark immediately after issuance and as deliverables into the Bund and Bobl future contracts. In this way, the fair value of the repo–GC basis swap